Saketh Velidimalla

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Applied AI systems and data engineer with production ownership of evaluation, observability, analytics, and governed data layers for an enterprise RAG platform serving approximately 140,000 users. Builds measurable quality infrastructure, turns fragmented operational evidence into reliable data products, and translates findings into engineering and business decisions.

Core Competencies

- **Applied AI Systems:** RAG pipelines, LLM integration, retrieval and vector search, FastAPI, Docker, AWS Bedrock, LLM classification
- **Evaluation & Observability:** LLM-as-a-Judge, regression testing, Arize Phoenix, trace analysis, hallucination and retrieval-quality analysis, red teaming
- **Data Products & Engineering:** Python, SQL, PySpark, Airflow DAGs, AWS Glue, DynamoDB, Databricks, Delta/Parquet, AWS S3/EMR/Redshift, schema validation
- **Enterprise Analytics & Governance:** Power BI/DAX, semantic models, ServiceNow analytics, IAM and OAuth access, metric lineage, PII classification, runbooks

Professional Experience

Johnson & Johnson — **New Brunswick, NJ**
Data Scientist — *Feb 2025–Present*

- Owned evaluation, analytics, reporting, and observability for a production enterprise RAG system serving approximately 140,000 internal users across knowledge retrieval, case-management, and content-governance workflows
- Built a three-layer LLM-as-judge pipeline on AWS Bedrock / Llama 3.2 combining rule-based classifiers, batch LLM categorization, and keyword fallbacks to evaluate thousands of queries daily
- Built LLM regression tests with ground-truth benchmarks, judge-based similarity scoring, and cross-run tracking to detect relevance, hallucination, latency, and retrieval-quality regressions before deployment
- Designed containment, query-resolution, repeat-user, and satisfaction metrics; analysis-guided content, formatting, and retrieval fixes raised response and containment rates by 10 percentage points
- Connected enterprise RAG traces, DynamoDB records, API payloads, and ServiceNow workflow tables to separate AI, routing, knowledge, and downstream-intake failure modes with row-level evidence
- Built ETL pipelines, including AWS Glue jobs that generate Parquet for automatic full and delta ingestion into Databricks, and Power BI semantic models with KPI definitions and lineage notes
- Wrote guardrails and red-teamed the assistant; set up read-only production-table access through IAM roles and OAuth, and role-based report access and filtering through Power BI service roles
- Derived an organization flag from a ServiceNow access rule that filters KPIs and dashboards, switches the prompt library per organization, and sets organization rules for raising cases on a user's behalf

Hewlett Packard Inc. — **Spring, TX**
Data Science Engineer — *Jul 2024–Nov 2024*

- Built Airflow DAGs running PySpark transformations from S3 PC telemetry into Redshift views for Power BI reporting across hardware, software, power, sensor, and BIOS data
- Optimized PySpark processing on AWS EMR, reducing runtime by approximately 25% through query optimization, repartitioning, and pipeline tuning
- Built engineering KPI reports that helped hardware, software, and chip-design stakeholders compare telemetry trends, prioritize defects, and support roadmap and funding decisions
- Translated raw hardware telemetry into stable analytical tables and Power BI measures so engineering teams could inspect component-level trends without relying on ad-hoc extracts

ECrent Worldwide Company — **Remote**
Machine Learning Engineer — *Jul 2021–Jun 2022*

- Built and deployed a content-based recommendation system for real-estate listings using structured attributes and text descriptions to generate personalized suggestions
- Implemented semantic-similarity ranking with Word2Vec embeddings and evaluated results with Precision-at-K and Recall-at-K
- Integrated top-N retrieval into the application backend for real-time Flask API serving
- Prepared feature extraction and evaluation scripts that made recommendation changes measurable before exposing ranked listings through the application workflow

Selected Systems Work

- **SakethWiki:** Local RAG reliability system with curated ingestion, review-gated amendments, telemetry, trace review, evaluation replay, regression gates, and task-based model routing.
- **roastrank-v2:** LLM evaluation and benchmarking platform with rubric-based scoring, regression testing, six-stage output validation, failure labeling, and audit logging. FastAPI, Docker, Hugging Face.
- **SDLC Intelligence Agent:** Built a LangGraph multi-agent workflow that summarizes CI failures, maps errors to source files, flags risky changes, and produces reviewer guidance.

Education & Development

Arizona State University — **Phoenix, AZ**
MS Information Technology (Data Science) — *Aug 2022–May 2024*

Completed: Claude Code in Action, Claude 101, Building with Claude API, Agentic AI specialization (Andrew Ng / DeepLearning.AI)
AI tooling: Codex, Claude Code **In progress:** LangGraph Fundamentals, LLM Engineering